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Here's a **comprehensive checklist** to perform an **in-place upgrade of SQL Server 2016 Developer Edition to SQL Server 2019 Developer Edition**. An in-place upgrade means upgrading the existing SQL Server instance without moving databases to a different server or instance.

✅ SQL Server In-Place Upgrade Checklist (2016 ➜ 2019)

1. 📋 Pre-Upgrade Planning

- **Review Microsoft Documentation**
  - Confirm that in-place upgrade from SQL Server 2016 ➜ 2019 is supported.
- **Check SQL Server 2019 System Requirements**
  - OS compatibility (Windows Server version, disk space, memory, CPU).
  - .NET Framework version (4.6 or higher).
- **Review Deprecated Features**
  - Use SQL Server Upgrade Advisor or Data Migration Assistant (DMA) to identify deprecated/unsupported features.
- **Check Licensing**
  - Ensure Developer Edition is being used (free but not for production).

2. 💾 Backups & Recovery Preparation

- **Full Backup of All Databases**
  - Include system databases: master, msdb, model, and msdb.
- **Backup SQL Agent Jobs, Logins, Linked Servers**
  - Use scripts or SSMS export.
- **Document Configuration Settings**
  - Server-level settings, custom configurations, SQL Server configuration manager settings.
- **Backup SQL Server Configuration Files**
  - sqlserver.ini, error logs, etc.
- **Check Disaster Recovery Plan**
  - Verify ability to roll back to 2016 if upgrade fails (snapshot, VM backup, etc.).

3. 🔎 Assessment Tools

- **Run Data Migration Assistant (DMA)**
  - Analyze issues/blockers, deprecated features, and compatibility.
- **Check for Compatibility Level**
  - Document the compatibility level of each database (SELECT compatibility_level FROM sys.databases).
- **Check Database Integrity**
  - Run DBCC CHECKDB on all databases.
- **Free Up Space**
  - Ensure sufficient disk space for upgrade (at least 20–30 GB free depending on database size).

4. 🔄 Perform the Upgrade

- **Download SQL Server 2019 Developer Edition**
  - From Microsoft official website.
- **Run SQL Server 2019 Setup**
  - Choose **Upgrade from a previous version of SQL Server**.
- **Select Instance to Upgrade**
  - Choose the existing SQL Server 2016 instance.
- **Follow Setup Wizard**

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  - Validate setup support rules.
  - Accept license terms.
  - Confirm instance and features.
- **Install Required Updates**
  - Let the installer check for and apply SQL Server 2019 updates.
- **Finish Upgrade and Reboot (if prompted)**

5. ✅ Post-Upgrade Tasks

- **Check SQL Server Services**
  - Ensure all services are running (SQL Server, Agent, SSAS, etc.).
- **Check SQL Server Version**
  - Run SELECT @@VERSION; to confirm upgrade.
- **Run Application Tests**
  - Validate connectivity and functionality of all apps using SQL Server.
- **Reconfigure Maintenance Plans/Jobs**
  - Ensure Agent jobs and schedules are working.
- **Check Database Compatibility Levels**
  - Optionally raise compatibility level to 150 (SQL Server 2019) after testing.
- **Update Statistics and Rebuild Indexes**
  - Recommended after major upgrades.
- **Enable Query Store** (Optional)
  - Useful for performance monitoring and regressions.

6. 📘 Documentation & Monitoring

- **Document the Upgrade Process & Outcomes**
- **Set Up Monitoring & Alerts**
  - Ensure monitoring tools are compatible and working.
- **Check Logs for Errors**
  - Review SQL Server logs and Windows Event Logs.
- **Install Latest Cumulative Updates (CU) for SQL Server 2019**

🛑 Rollback Plan (Important)

- Have **full VM/server image backup** or be ready to **reinstall SQL Server 2016** and restore backups.
- Make sure you can recover all DBs, jobs, logins, and configs if upgrade fails.